



Lecture # 6

12 Principles of Project Management (PMBOK 7th Edition)

ENGINEERING PROJECT MANAGEMENT

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1: Stewardship

3.1 BE A DILIGENT, RESPECTFUL, AND CARING STEWARD

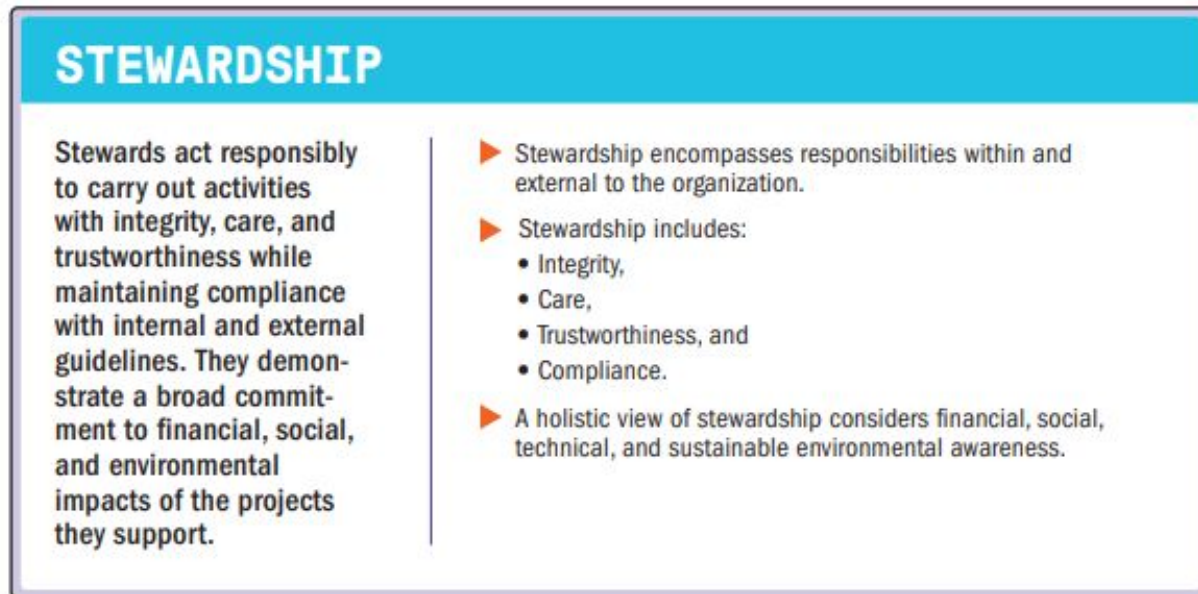


Figure 3-2. Be a Diligent, Respectful, and Caring Steward

2. Team

3.2 CREATE A COLLABORATIVE PROJECT TEAM ENVIRONMENT

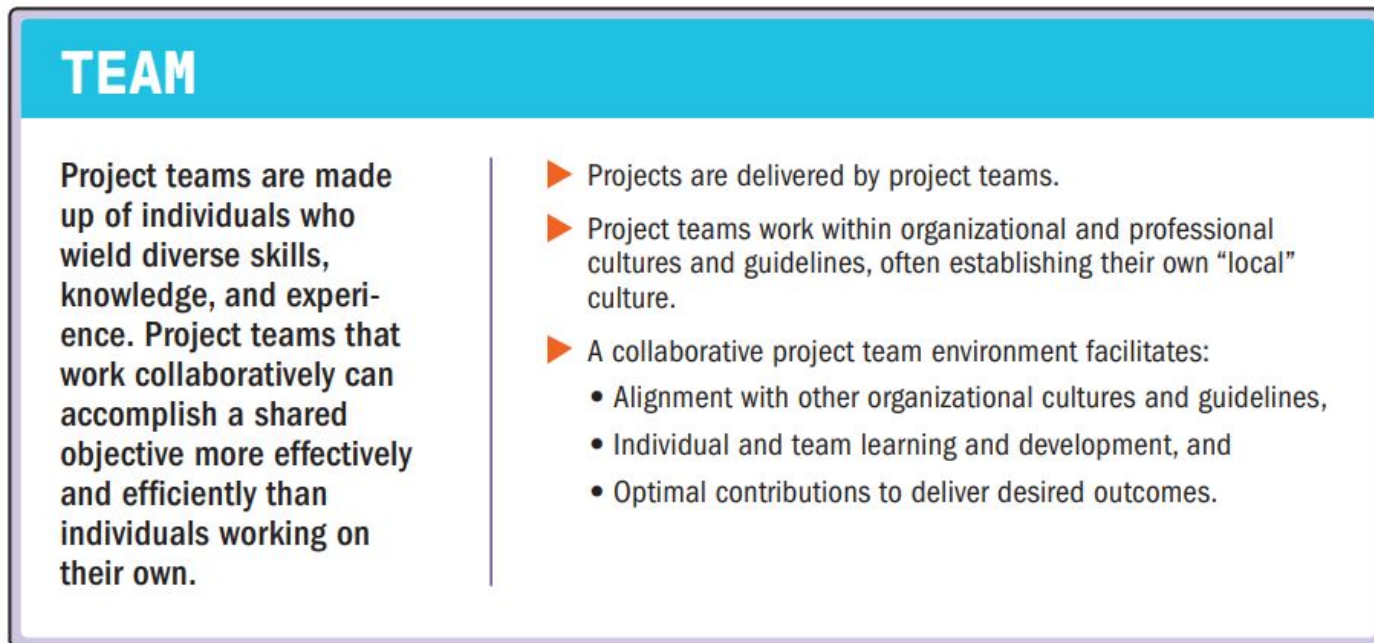


Figure 3-3. Create a Collaborative Project Team Environment

3. Stakeholder

3.3 EFFECTIVELY ENGAGE WITH STAKEHOLDERS

STAKEHOLDERS

Engage stakeholders proactively and to the degree needed to contribute to project success and customer satisfaction.

- ▶ Stakeholders influence projects, performance, and outcomes.
- ▶ Project teams serve other stakeholders by engaging with them.
- ▶ Stakeholder engagement proactively advances value delivery.

Figure 3-4. Effectively Engage with Stakeholders

3. Stakeholder

Stakeholders can affect many aspects of a project, including but not limited to:

- ▶ ▶ Scope/requirements, by revealing the need to add, adjust, or remove elements of the scope and/or project requirements;
- ▶ ▶ Schedule, by offering ideas to accelerate delivery or by slowing down or stop delivery of key project activities;
- ▶ ▶ Cost, by helping to reduce or eliminate planned expenditures or by adding steps, requirements, or restrictions that increase cost or require additional resources;
- ▶ ▶ Project team, by restricting or enabling access to people with the skills, knowledge, and experience needed to deliver the intended outcomes, and promote a learning culture;
- ▶ ▶ Plans, by providing information for plans or by advocating for changes to agreed activities and work;
- ▶ ▶ Outcomes, by enabling or blocking work required for the desired outcomes;
- ▶ ▶ Culture, by establishing or influencing—or even defining—the level and character of engagement of the project team and broader organization;
- ▶ ▶ Benefits realization, by generating and identifying long-term goals so that the project delivers the intended identified value;
- ▶ ▶ Risk, by defining the risk thresholds of the project, as well as participating in subsequent risk management activities;
- ▶ ▶ Quality, by identifying and requiring quality requirements; and
- ▶ ▶ Success, by defining success factors and participating in the evaluation of success.

4. Value

3.4 FOCUS ON VALUE

VALUE

Continually evaluate and adjust project alignment to business objectives and intended benefits and value.

- ▶ Value is the ultimate indicator of project success.
- ▶ Value can be realized throughout the project, at the end of the project, or after the project is complete.
- ▶ Value, and the benefits that contribute to value, can be defined in quantitative and/or qualitative terms.
- ▶ A focus on outcomes allows project teams to support the intended benefits that lead to value creation.
- ▶ Project teams evaluate progress and adapt to maximize the expected value.

Figure 3-5. Focus on Value

5. System Thinking

3.5 RECOGNIZE, EVALUATE, AND RESPOND TO SYSTEM INTERACTIONS

SYSTEMS THINKING

Recognize, evaluate, and respond to the dynamic circumstances within and surrounding the project in a holistic way to positively affect project performance.

- ▶ A project is a system of interdependent and interacting domains of activity.
- ▶ Systems thinking entails taking a holistic view of how project parts interact with each other and with external systems.
- ▶ Systems are constantly changing, requiring consistent attention to internal and external conditions.
- ▶ Being responsive to system interactions allows project teams to leverage positive outcomes.

Figure 3-6. Recognize, Evaluate, and Respond to System Interactions

6. Leadership

3.6 DEMONSTRATE LEADERSHIP BEHAVIORS

LEADERSHIP

Demonstrate and adapt leadership behaviors to support individual and team needs.

- ▶ Effective leadership promotes project success and contributes to positive project outcomes.
- ▶ Any project team member can demonstrate leadership behaviors.
- ▶ Leadership is different than authority.
- ▶ Effective leaders adapt their style to the situation.
- ▶ Effective leaders recognize differences in motivation among project team members.
- ▶ Leaders demonstrate desired behavior in areas of honesty, integrity, and ethical conduct.

Figure 3-7. Demonstrate Leadership Behaviors

7. Tailoring

3.7 TAILOR BASED ON CONTEXT

TAILORING

Design the project development approach based on the context of the project, its objectives, stakeholders, governance, and the environment using “just enough” process to achieve the desired outcome while maximizing value, managing cost, and enhancing speed.

- ▶ Each project is unique.
- ▶ Project success is based on adapting to the unique context of the project to determine the most appropriate methods of producing the desired outcomes.
- ▶ Tailoring the approach is iterative, and therefore is a continuous process throughout the project.

Figure 3-8. Tailor Based on Context

8. Quality

3.8 BUILD QUALITY INTO PROCESSES AND DELIVERABLES

QUALITY

Maintain a focus on quality that produces deliverables that meet project objectives and align to the needs, uses, and acceptance requirements set forth by relevant stakeholders.

- ▶ Project quality entails satisfying stakeholders' expectations and fulfilling project and product requirements.
- ▶ Quality focuses on meeting acceptance criteria for deliverables.
- ▶ Project quality entails ensuring project processes are appropriate and as effective as possible.

Figure 3-9. Build Quality into Processes and Deliverables

8. Quality

Quality may have several different dimensions, including but not limited to the following:

- ▶ **Performance**. Does the deliverable function as the project team and other stakeholders intended?
- ▶ **Conformity**. Is the deliverable fit for use, and does it meet the specifications?
- ▶ **Reliability**. Does the deliverable produce consistent metrics each time it is performed or produced?
- ▶ **Resilience**. Is the deliverable able to cope with unforeseen failures and quickly recover?
- ▶ **Satisfaction**. Does the deliverable elicit positive feedback from end users? This includes usability and user experience?
- ▶ **Uniformity**. Does the deliverable show parity with other deliverables produced in the same manner?
- ▶ **Efficiency**. Does the deliverable produce the greatest output with the least amount of inputs and effort?
- ▶ **Sustainability**. Does the deliverable produce a positive impact on economic, social, and environmental parameters

9. Complexity

3.9 NAVIGATE COMPLEXITY

COMPLEXITY

Continually evaluate and navigate project complexity so that approaches and plans enable the project team to successfully navigate the project life cycle.

- ▶ Complexity is the result of human behavior, system interactions, uncertainty, and ambiguity.
- ▶ Complexity can emerge at any point during the project.
- ▶ Complexity can be introduced by events or conditions that affect value, scope, communications, stakeholders, risk, and technological innovation.
- ▶ Project teams can stay vigilant in identifying elements of complexity and use a variety of methods to reduce the amount or impact of complexity.

Figure 3-10. Navigate Complexity

10. Risk

3.10 OPTIMIZE RISK RESPONSES

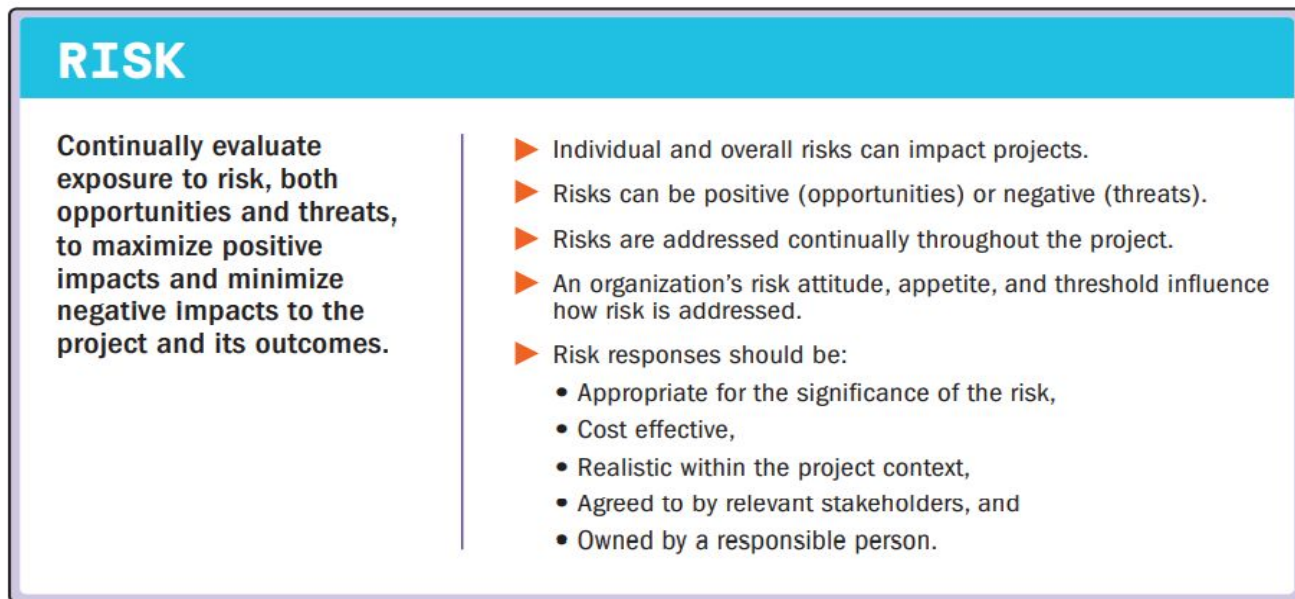


Figure 3-11. Optimize Risk Responses

11. Adaptability

3.11 EMBRACE ADAPTABILITY AND RESILIENCY

ADAPTABILITY AND RESILIENCY

Build adaptability and resiliency into the organization's and project team's approaches to help the project accommodate change, recover from setbacks, and advance the work of the project.

- ▶ Adaptability is the ability to respond to changing conditions.
- ▶ Resiliency is the ability to absorb impacts and to recover quickly from a setback or failure.
- ▶ A focus on outcomes rather than outputs facilitates adaptability.

Figure 3-12. Embrace Adaptability and Resiliency

12. Change

3.12 ENABLE CHANGE TO ACHIEVE THE ENVISIONED FUTURE STATE

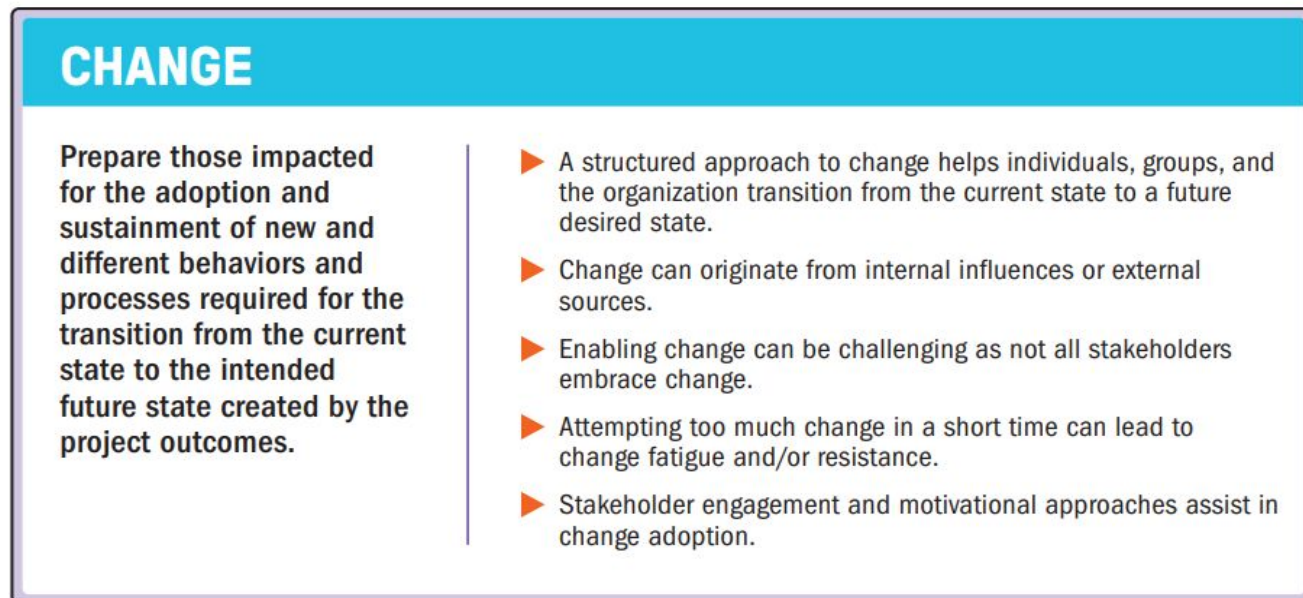


Figure 3-13. Enable Change to Achieve the Envisioned Future State

Principles of Project Management

Be a diligent, respectful, and caring steward	Create a collaborative team environment	Effectively engage with stakeholders	Focus on value
Recognize, evaluate, and respond to system interactions	Demonstrate leadership behaviors	Tailor based on context	Build quality into processes and deliverables
Navigate complexity	Optimize risk responses	Embrace adaptability and resiliency	Enable change to achieve the envisioned future state

Guide Behavior



Discussion

